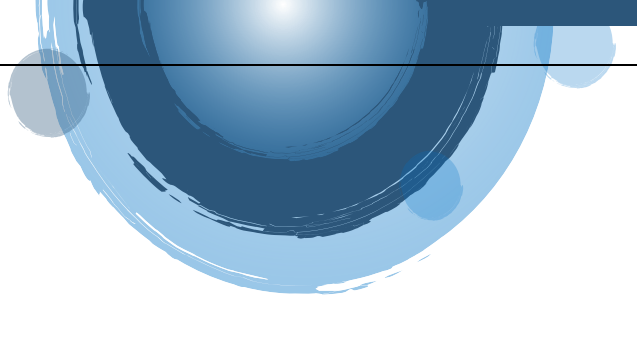




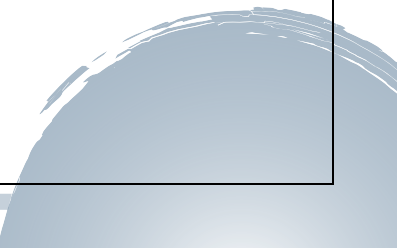
# Cyber Security Internship Report

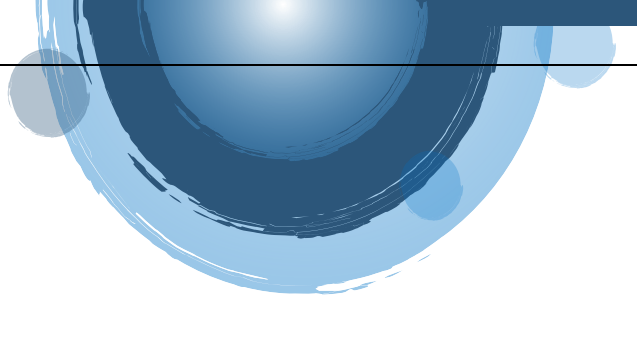
Cyber Security Internship – Task 14

## **Linux Server Hardening & Secure Configuration Report**

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Date : 10 / 02 / 2026





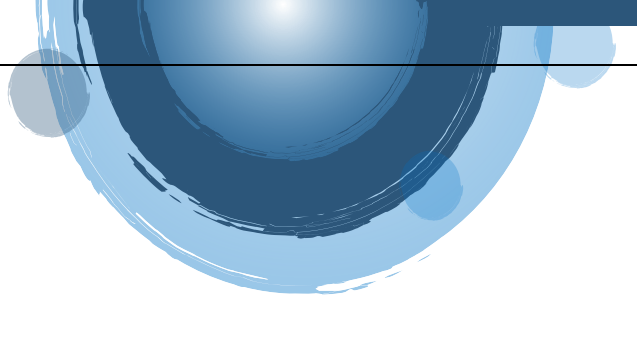
## 1. Introduction

Linux servers are widely used in enterprise environments to host web applications, databases, and critical services. Due to their exposure to public networks, they are common targets for cyberattacks such as brute-force attacks, privilege escalation, malware installation, and unauthorized access.

Default Linux installations often contain unnecessary services, open ports, and weak configurations that increase the attack surface. Therefore, server hardening is essential to improve system security and reduce vulnerabilities.

This task focuses on applying industry-standard hardening techniques to secure a Linux server against common threats and misconfigurations.

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## 2. Objectives

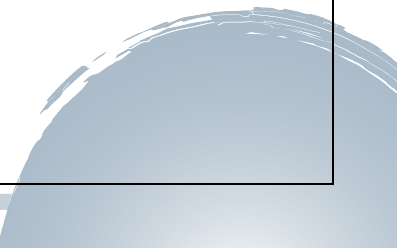
The primary objectives of this task are:

- Reduce attack surface
- Remove unnecessary services and users
- Strengthen authentication mechanisms
- Secure network communication
- Enforce least privilege access
- Enable monitoring and logging
- Audit system security posture

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## 3. Tools Used

| Tool                | Purpose                   |
|---------------------|---------------------------|
| Kali Linux / Ubuntu | Server platform           |
| SSH                 | Secure remote access      |
| UFW Firewall        | Network traffic filtering |
| Lynis               | Security auditing         |





## Tool

journalctl

unattended-  
upgrades

## Purpose

Log monitoring

Automatic security  
patches

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## 4. Methodology / Implementation Steps

### 4.1 System Update & Patch Management

Outdated packages may contain known vulnerabilities. Therefore, updating the system is the first step.

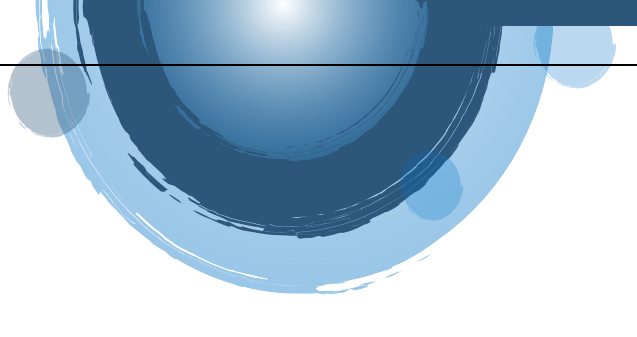
#### Commands:

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt autoremove -y
```

#### Result:

- Installed latest security patches
  - Removed obsolete packages
-



## 4.2 User & Access Management

Multiple unnecessary users increase risk of unauthorized access.

### **Actions performed:**

- Reviewed all system users
- Removed unused accounts
- Restricted sudo privileges

### **Commands:**

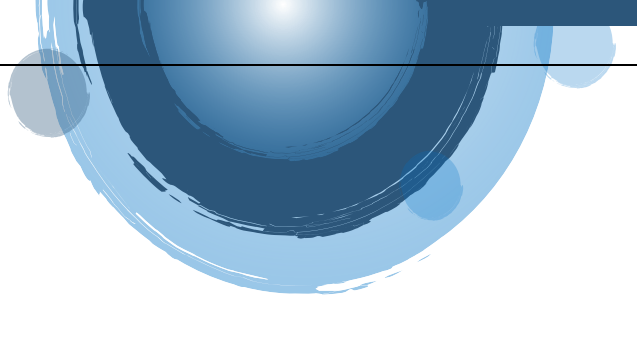
```
cat /etc/passwd
```

```
sudo deluser username
```

```
sudo usermod -aG sudo hetal
```

### **Security Benefit:**

- Reduced unauthorized access points
  - Implemented least privilege principle
-



## 4.3 SSH Hardening

SSH is a common attack target. Default settings allow password-based attacks.

### Changes made:

- Disabled root login
- Disabled password authentication
- Enabled key-based authentication

### Commands:

```
sudo nano /etc/ssh/sshd_config
```

```
PermitRootLogin no
```

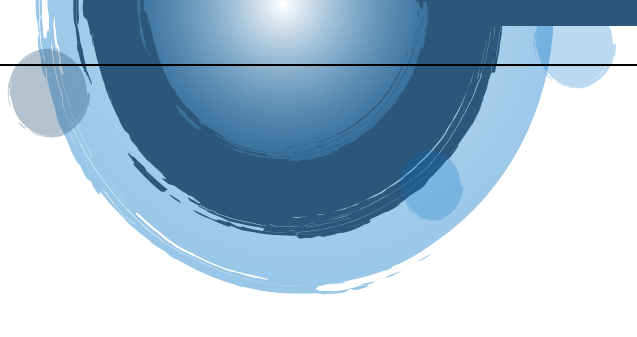
```
PasswordAuthentication no
```

```
sudo systemctl restart ssh
```

```
ssh-keygen -t rsa -b 4096
```

### Security Benefit:

- Prevents brute-force attacks
  - Strong cryptographic authentication
-



## 4.4 Firewall Configuration

Firewall controls incoming/outgoing traffic.

### **UFW configuration:**

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

```
sudo ufw allow 22
```

```
sudo ufw enable
```

### **Security Benefit:**

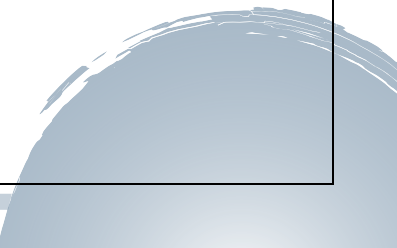
- Blocks unauthorized ports
- Minimizes exposure to attackers

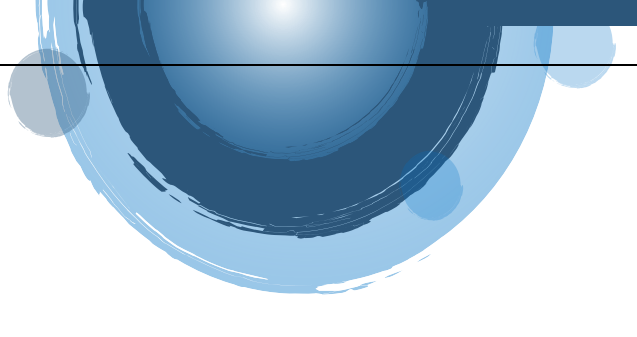
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## 4.5 Service Hardening

Unnecessary services increase vulnerability.

### **Steps:**

- Listed enabled services
  - Disabled unused ones
- 



## Commands:

```
systemctl list-unit-files --type=service --  
state=enabled
```

```
sudo systemctl disable bluetooth
```

## Security Benefit:

- Reduced attack surface
  - Improved performance
- 

## 4.6 File Permission Security

Sensitive files must be protected.

## Commands:

```
sudo chmod 700 /root
```

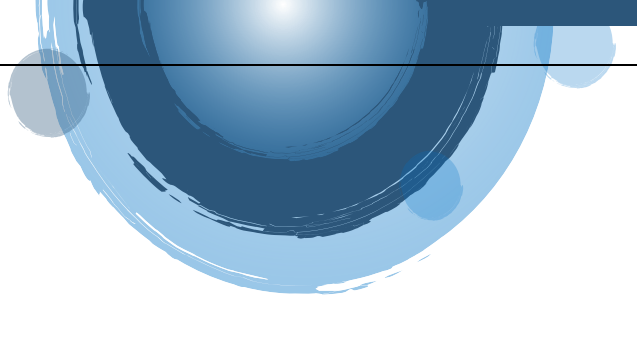
```
sudo chmod 600 /etc/shadow
```

```
sudo chmod 600 ~/.ssh/authorized_keys
```

## Security Benefit:

- Prevents unauthorized access to credentials
-





## 4.7 Automatic Security Updates

Ensures future vulnerabilities are patched automatically.

### **Commands:**

```
sudo apt install unattended-upgrades
```

```
sudo dpkg-reconfigure unattended-upgrades
```

### **Security Benefit:**

- Continuous protection

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## 4.8 Security Auditing with Lynis

Performed vulnerability scan.

### **Command:**

```
sudo lynis audit system
```

### **Outcome:**

- Identified misconfigurations
  - Provided hardening recommendations
-

## 4.9 Log Monitoring

Logs help detect suspicious activity.

### Commands (Kali specific):

```
sudo journalctl -f
```

```
sudo journalctl -u ssh
```

```
sudo journalctl | grep Failed
```

### Security Benefit:

- Detects brute-force attempts
- Tracks authentication failures

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## 5. Linux Hardening Checklist

| Security Control     | Status |
|----------------------|--------|
| System updated       | ✓      |
| Users reviewed       | ✓      |
| Unused users removed | ✓      |
| Root login disabled  | ✓      |

## Security Control

## Status

SSH key authentication



Firewall enabled



Unused services disabled



File permissions secured



Automatic updates enabled



Logs monitored



Security audit completed

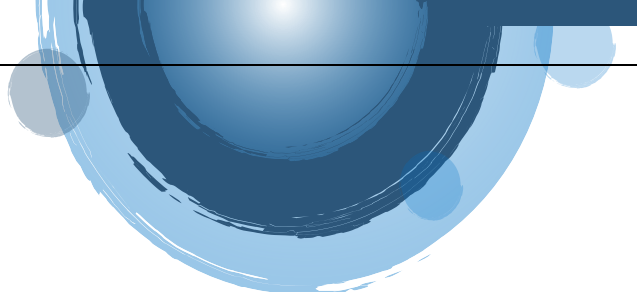


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## 6. Security Configuration Summary

After applying all hardening techniques:

- Only required services are running
- Unauthorized ports blocked
- Strong SSH authentication enforced
- Root access restricted
- Regular updates enabled
- Continuous monitoring active



This significantly reduces risks including:

- Brute force attacks
- Unauthorized login
- Privilege escalation
- Malware exploitation
- Service abuse

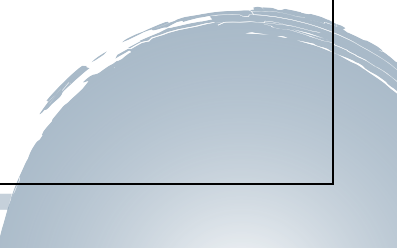
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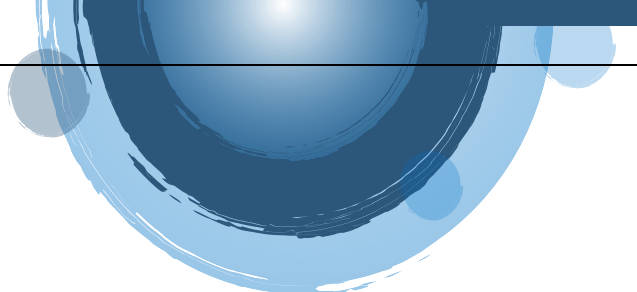
## 7. Results / Outcome

### **Before Hardening:**

- Multiple open services
- Password login enabled
- No firewall
- No monitoring

### **After Hardening:**

- Minimal services
  - Secure SSH configuration
- 

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- Active firewall
  - Continuous updates
  - Log monitoring
  - Reduced attack surface

The server is now secure and production-ready.

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## **8. Challenges Faced**

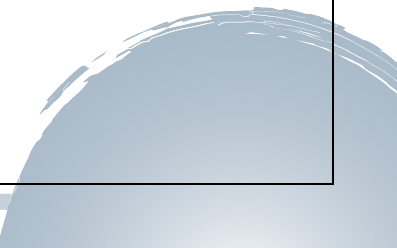
- apt lock issue during package installation
- Kali Linux uses journalctl instead of auth.log
- Service identification for disabling

These were resolved through troubleshooting and research.

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## **9. Learning Outcomes**

Through this task, I learned:

- Linux system administration
  - Security best practices
- 

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- SSH hardening
  - Firewall configuration
  - Log analysis
  - Vulnerability scanning
  - Server attack surface reduction

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## 10. Conclusion

Linux server hardening is a critical cybersecurity practice. By applying systematic configuration changes, removing unnecessary components, and enforcing strict access controls, the system becomes significantly more secure.

This task successfully demonstrated how proper hardening techniques can protect Linux servers from common threats and real-world attacks.

